

DSC 40A

Theoretical Foundations of Data Science I

Lecture 20-21: Combinatorics

Agenda

- How do we count the number of outcomes, besides enumerating them all?
 - How many outcomes are possible if a die is rolled 100 times?
 - How many different ways are there to shuffle 52 cards?
 - How many ways are there to choose a jury of 12 people from a panel of 100?
- Many probability questions can be solved by counting, or combinatorics.
- We'll learn how to count sequences and sets.

Question

Answer at q.dsc40a.com

Remember, you can always ask questions at
q.dsc40a.com!

If the direct link doesn't work, click the "Lecture Questions" link in the top right corner of dsc40a.com.

Combinatorics

The background of the slide features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green. These shapes are concentrated on the right side and bottom, creating a modern, geometric aesthetic. The word "Combinatorics" is written in a green, sans-serif font on the left side of the slide.

Today

- How do we count the number of outcomes, besides enumerating them all?
 - How many outcomes are possible if a die is rolled 100 times?
 - How many different ways are there to shuffle 52 cards?
 - How many ways are there to choose a jury of 12 people from a panel of 100?
- Many probability questions can be solved by counting, or combinatorics.
- We'll learn how to count sequences and sets.

Sequences vs. Sets

Sequences	Sets
Order matters	Order does not matter
Repetitions allowed (with replacement)	No repetitions allowed (without replacement)
Elements listed in order	Elements listed in no particular order within curly braces
Ex: $2, 4, 5 \neq 4, 2, 5$	Ex: $\{2, 4, 5\} = \{4, 2, 5\}$
Ex: $2, 2, 2 \neq 2, 2$	Ex: $\{2, 2, 2\} = \{2, 2\} = \{2\}$
Ex: $1, 3, 4 = 1, 3, 4$	Ex: $\{1, 3, 4\} = \{1, 3, 4\}$

Sequences

Sequences

Order matters

Repetitions allowed

Elements listed in order

Ex: $2, 4, 5 \neq 4, 2, 5$

Ex: $2, 2, 2 \neq 2, 2$

Ex: $1, 3, 4 = 1, 3, 4$

Example 1:

draw a card, put it back, repeat four more times

$(A\heartsuit, 2\clubsuit, 6\spadesuit, A\heartsuit, 3\diamondsuit)$

Example 2:

flip a coin 100 times

$(H, T, T, H, \dots, H, T, T, T)$

Sequences

Sequences

Order matters

Repetitions allowed

Elements listed in order

Ex: $2, 4, 5 \neq 4, 2, 5$

Ex: $2, 2, 2 \neq 2, 2$

Ex: $1, 3, 4 = 1, 3, 4$

A UCSD PID starts with “A” then has 8 digits.
How many UCSD PIDs are possible?

A. 8^{10}

C. $8!$

B. 10^8

D.

Sequences

Sequences

Order matters

Repetitions allowed

Elements listed in order

Ex: $2, 4, 5 \neq 4, 2, 5$

Ex: $2, 2, 2 \neq 2, 2$

Ex: $1, 3, 4 = 1, 3, 4$

A UCSD PID starts with “A” then has 8 digits.
How many UCSD PIDs are possible?

A. 8^{10}

C. $8!$

B. 10^8

D.

P is the population you can draw from and $|P|$ is the size of that population (number of elements).

How many sequences of length k are there?

$$\underbrace{|P| \cdot |P| \cdots |P|}_{k \text{ times}} = |P|^k$$

Sequences

Sequences
Order matters
Repetitions allowed
Elements listed in order
Ex: $2, 4, 5 \neq 4, 2, 5$
Ex: $2, 2, 2 \neq 2, 2$
Ex: $1, 3, 4 = 1, 3, 4$

Exponential growth
Flip a coin n times

n	# of Sequences of Length n
5	$2^5 = 32$
10	$2^{10} = 1024$
15	$2^{15} = 32,768$
20	$2^{20} \approx 1 \text{ million}$
50	$2^{50} \approx \# \text{ of grains of sand on Earth}$

Sequences

Sequences

Order matters

Repetitions allowed

Elements listed in order

Ex: $2, 4, 5 \neq 4, 2, 5$

Ex: $2, 2, 2 \neq 2, 2$

Ex: $1, 3, 4 = 1, 3, 4$

How many ways to select a president, vice president, and secretary from a group of 8 people?

Sequences

Sequences

Order matters

Repetitions allowed

Elements listed in order

Ex: $2, 4, 5 \neq 4, 2, 5$

Ex: $2, 2, 2 \neq 2, 2$

Ex: $1, 3, 4 = 1, 3, 4$

How many ways to select a president, vice president, and secretary from a group of 8 people?

Sequences where repetitions are not allowed are _____.

Sets

There are 24 ice cream flavors. How many ways can you pick 2 different flavors?

A. 24

C. 24×24

B. 24×23

D. 12×23

Sets

Order does not matter

No repetitions allowed

Elements listed in no particular order within curly braces

Ex: $\{2, 4, 5\} = \{4, 2, 5\}$

Ex: $\{2, 2, 2\} = \{2, 2\} = \{2\}$

Ex: $\{1, 3, 4\} = \{1, 3, 4\}$

Sets

How many ways to select a committee of 3 from a group of 8?

Sets

Order does not matter

No repetitions allowed

Elements listed in no particular order within curly braces

Ex: $\{2, 4, 5\} = \{4, 2, 5\}$

Ex: $\{2, 2, 2\} = \{2, 2\} = \{2\}$

Ex: $\{1, 3, 4\} = \{1, 3, 4\}$

Permutations vs. Combinations

Permutations	Combinations
Order matters	Order does not matter
No repetitions allowed (without replacement)	No repetitions allowed (without replacement)
Counts the number of sequences of k distinct elements chosen from n possible elements	Counts the number of sets of size k chosen from n possible elements
$P(n, k) = (n)(n - 1) \dots (n - k + 1) = \frac{n!}{(n - k)!}$	$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$
How many ways to select a president, vice president, and secretary from a group of 8 people? $P(8, 3)$	How many ways to select a committee of 3 from a group of 8? $C(8, 3)$

Permutations vs. Combinations

Permutations

Order matters

No repetitions allowed (without replacement)

Counts the number of **sequences of k distinct elements** chosen from n possible elements

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

How many ways to select a president, vice president, and secretary from a group of 8 people?
 $P(8,3)$

Example 1:

draw a card, **don't** put it back, repeat four more times

$(A♥, 2♣, 6♠, 7♥, 3♦)$

Example 2:

rank 2 best cities to live in out of list of 10

SD, LA

Permutations vs. Combinations

Combinations

Order does not matter

No repetitions allowed (without replacement)

Counts the number of **sets of size k** chosen from n possible elements

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

How many ways to select a committee of 3 from a group of 8?

$$C(8, 3)$$

Example 1:

draw a hand of 5 cards from a deck of 52

(A♥, 2♣, 6♠, 7♥, 3♦)

Example 2:

Select 5 student from the class

Sampling Without Replacement

Example 6. There are 20 students in a class. A computer program selects a random sample of students by drawing 5 students at random **without replacement**. What is the chance that a particular student is among the 5 selected students?

Sampling Without Replacement

Part 1. Denominator. If you draw a sample of size 5 at random without replacement from a population of size 20, how many different **sets** of individuals could you draw?

Sampling Without Replacement

Part 2. Numerator. If you draw a sample of size 5 at random without replacement from a population of size 20, how many different **sets** of individuals include a particular person?

Sampling Without Replacement

Using the complement. If you draw a sample of size 5 at random without replacement from a population of size 20, how many different **sets** of individuals **do not** include a particular person?

Sampling Without Replacement

Example 6. There are 20 students in a class. A computer program selects a random sample of students by drawing 5 students at random **without replacement**. What is the chance that a particular student is among the 5 selected students?

Summary

- To calculate $P(E)$ we need to find $|E|$
- We need to count sequences or sets
- Must decide if order matters
- When elements are distinct: permutations vs. combinations

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- **Next time:** more examples

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Today

- More examples of using combinatorics to solve probability questions.

Counting as a Tool for Probability

Example 7. What is the probability that a randomly generated bitstring of length 10 contains an equal number of zeros and ones?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting as a Tool for Probability

Example 8. What is the probability that a randomly generated bitstring of length 10 is the string 0011001101?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting as a Tool for Probability

Example 9. What is the probability that a fair coin flipped 10 times turns up an equal number of heads and tails?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting as a Tool for Probability

Example 10. What is the probability that a fair coin flipped 10 times turns up HHTTHHTTHT?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting as a Tool for Probability

Example 11. What is the probability that a biased coin with $Prob(H) = \frac{1}{3}$ flipped 10 times turns up an equal number of heads and tails?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting as a Tool for Probability

Example 12. What is the probability that a biased coin with $Prob(H) = \frac{1}{3}$ flipped 10 times turns up HHTTHHTTHT?

$$P(n, k) = (n)(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

The Easy Way

Example 6. There are 20 students in a class. A computer program selects a random sample of students by drawing 5 students at random **without replacement**. What is the chance that a particular student is among the 5 selected students?

The Easy Way

Example 6. There are 20 students in a class. A computer program selects a random sample of students by drawing 5 students at random **without replacement**. What is the chance that a particular student is among the 5 selected students?

Another way to think of sampling without replacement:

1. randomly shuffle all 20 students
2. take the first 5

Practice Problems

Example 13. You were one of 238 individuals who reported for jury duty. If 54 of these people will be assigned to a courtroom, what is the probability that you get assigned to a courtroom?

Practice Problems

Example 13. You were one of 238 individuals who reported for jury duty. If 54 of these people will be assigned to a courtroom, what is the probability that you get assigned to a courtroom?

How many sets of 54 individuals include you?

A. $C(238, 54)$

C. $C(238, 53)$

B. $C(237, 54)$

D. $C(237, 53)$

Practice Problems

Example 14. You were one of 238 individuals who reported for jury duty. Suppose 28 of these individuals are doctors. If 54 of these people will be assigned to a courtroom, what is the probability that exactly 5 doctors get assigned to a courtroom?

Practice Problems

Example 15. What is the probability that your five-card poker hand is a straight?

Practice Problems

Example 16. Suppose you look at your first card as it is dealt, and you see that it is a Queen. What is the probability that your five-card hand is a straight?

Summary

- Counting is a useful tool for probability.
- Sometimes there's an easy way!
- **Next time:** Bayes Theorem